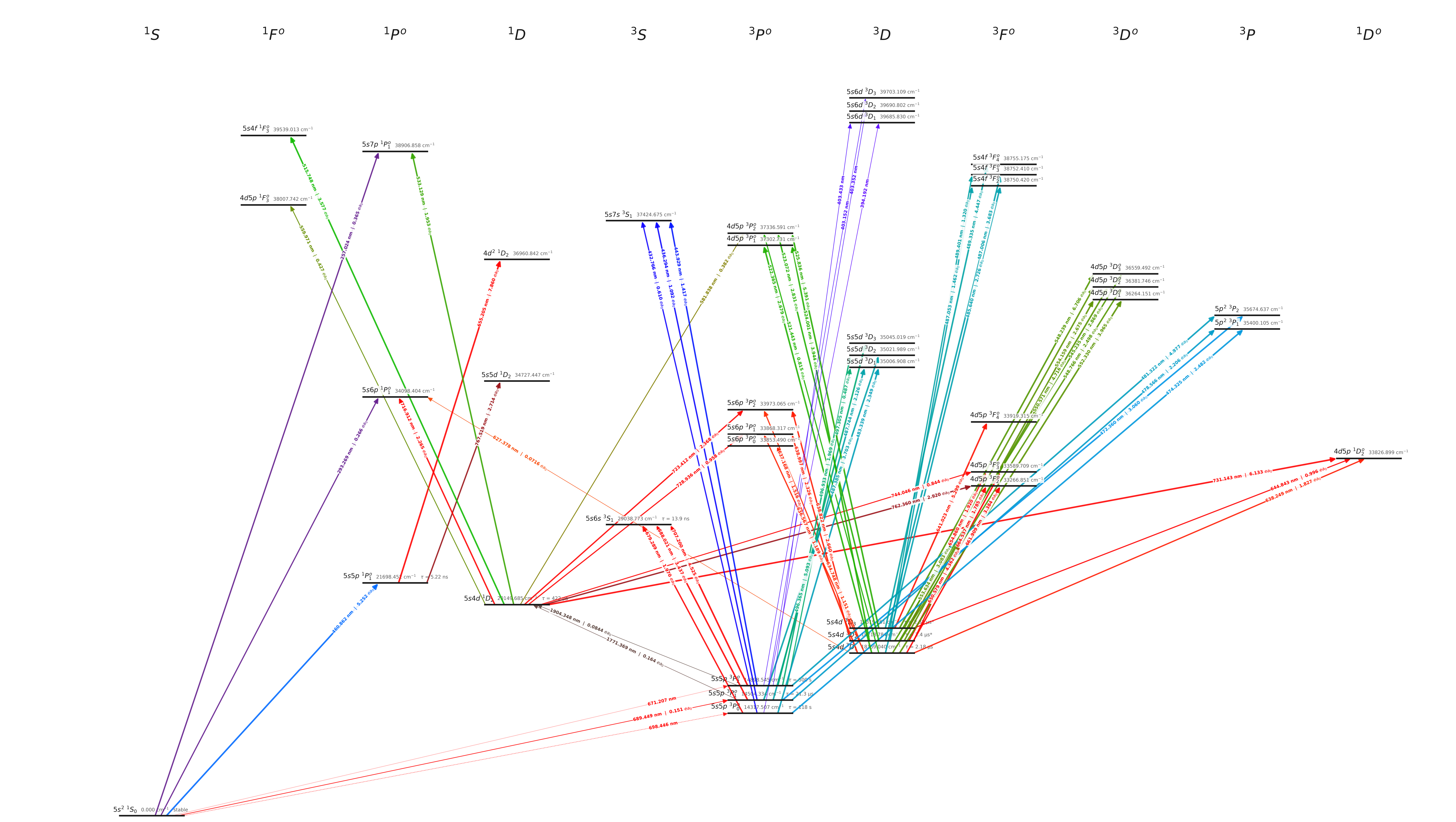


Strontium-87

⁸⁷Sr (Sr I), nuclear spin *I* = 9/2. Energy levels and transitions below 2 μm: 40 levels, 70 lines.

Sr⁺ ionisation limit 45932.20 cm⁻¹ = 5.69487 eV (217.712 nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	(J er J') (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
5s5p ³ P ₀ - 5s6d ³ D ₁	394.1924	394.0808	760.5232	-	-	-	- 394 nm	
5s5p ³ P ₂ - 5s6d ³ D ₃	403.1516	403.0377	743.6221	-	-	-	- 403.15 nm	
5s5p ³ P ₂ - 5s6d ³ D ₂	403.3517	403.2378	743.2532	-	-	-	- 403.35 nm repumper	
5s5p ³ P ₂ - 5s6d ³ D ₁	403.4326	403.3187	743.1041	-	-	-	- 403.43 nm	
5s ² ¹ S ₀ - 5s5p ¹ P ₁	460.8624	460.7333	650.5032	5.252	1.903e+08	30.51	100	461 nm blue MOT / Zeeman slower / imaging
5s5p ³ P ₀ - 5p ² ³ P ₁	474.3248	474.1922	632.0404	2.482	3.900e+07	-	-	- 474 nm
5s5p ¹ P ₁ - 5s5d ¹ D ₁	483.3393	483.2042	620.2526	2.349	3.300e+07	-	-	- 483 nm repumper
5s5p ³ P ₂ - 5s5d ³ D ₃	496.3648	496.2263	603.9761	5.093	6.140e+07	-	-	- 496.36 nm
5s5p ³ P ₂ - 5s5d ³ D ₂	496.9328	496.7942	603.2857	1.969	1.280e+07	-	-	- 496.93 nm repumper
5s5p ³ P ₂ - 5s5d ³ D ₁	497.3055	497.1668	602.8336	0.487	1.300e+06	-	-	- 497.30 nm
5s ² ¹ S ₀ - 5s5p ³ P ₂	671.2065	671.0212	446.6471	-	1.520e-04	3.183e-10	7.6	671 nm magnetic-quadrupole line
5s5p ³ P ₀ - 5s6s ³ S ₁	679.2894	679.1020	441.3324	1.970	8.360e+06	11.43	11.6	679 nm repumper
5s5p ³ P ₁ - 5s6s ³ S ₁	688.0211	687.8313	435.7315	3.437	2.450e+07	11.43	34.1	688 nm
5s ² ¹ S ₀ - 5s5p ³ P ₁	689.4491	689.2589	434.8290	0.151	4.700e+04	0.007479	100	689 nm red MOT
5s ² ¹ S ₀ - 5s5p ¹ P ₀	698.4457	698.2531	429.2280	-	-	1.349e-09	-	- 698 nm optical clock / optical qubit
5s5p ³ P ₂ - 5s6s ³ S ₁	707.1995	707.0046	423.91497(3)	4.525	3.910e+07	11.43	54.3	707 nm repumper
5s5p ³ P ₂ - 5s4d ¹ D ₂	1771.3690	1770.8854	169.2434	0.164	1.950e+03	0.0003772	82	1D2 decay returning atoms to the ground state via 3P1
5s5p ³ P ₂ - 5s4d ¹ D ₂	1904.3484	1903.8286	157.4252	0.084	4.180e+02	0.0003772	17.7	1D2 decay to the metastable reservoir state

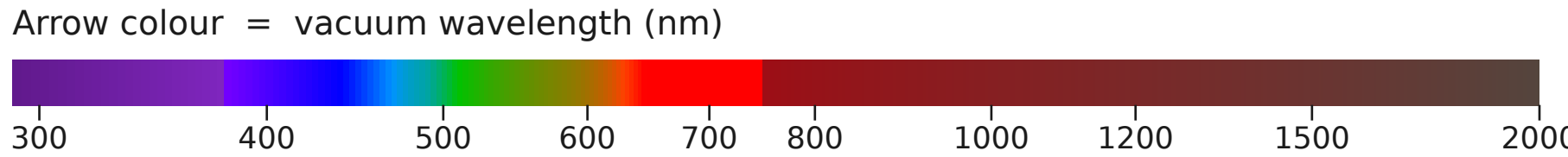
λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST. Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
5s5p ³ P ₀ - 5s6d ³ D ₁	394.192	88-84	91.05(4)
5s5p ¹ P ₀ - 5s6d ¹ D ₁	394.192	88-86	54.60(3)
5s5p ³ P ₂ - 5s6d ³ D ₁	394.192	88-87	51.641(28)
5s5p ³ P ₂ - 5s6d ³ D ₃	403.152	88-84	85(10)
5s5p ³ P ₂ - 5s6d ³ D ₃	403.152	88-86	41(10)
5s5p ³ P ₂ - 5s6d ³ D ₃	403.152	88-87	53(5)
5s5p ³ P ₂ - 5s6d ³ D ₂	403.352	88-84	81.0(20)
5s5p ³ P ₂ - 5s6d ³ D ₂	403.352	88-86	49.0(20)
5s5p ¹ P ₂ - 5s6d ¹ D ₂	403.352	88-87	20(5)
5s5p ³ P ₂ - 5s6d ³ D ₁	403.433	88-84	77(10)
5s5p ³ P ₂ - 5s6d ³ D ₁	403.433	88-86	30(10)
5s5p ³ P ₂ - 5s6d ³ D ₁	403.433	88-87	18.0(20)
5s ² ¹ S ₀ - 5s5p ¹ P ₁	460.862	84-88	-275(3)
5s ² ¹ S ₀ - 5s5p ¹ P ₁	460.862	86-88	-126.0(20)
5s ² ¹ S ₀ - 5s5p ¹ P ₁	460.862	87-88	-47.0(20)
5s5p ³ P ₀ - 5s5d ³ D ₁	483.339	88-84	90.00
5s5p ¹ P ₀ - 5s5d ¹ D ₁	483.339	88-86	40.00
5s5p ³ P ₀ - 5s5d ³ D ₁	483.339	88-87	32.00
5s5p ³ P ₂ - 5s5d ³ D ₃	496.365	88-84	91.60(20)
5s5p ³ P ₂ - 5s5d ³ D ₃	496.365	88-86	46.60(20)
5s5p ³ P ₂ - 5s5d ³ D ₃	496.365	88-87	27.0(20)
5s5p ³ P ₂ - 5s5d ³ D ₂	496.933	88-84	92.80(20)

Shift = ν(first isotope) - ν(second isotope).

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J || er || J' \rangle |$ in ea₀. Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation. Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70. Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



⁸⁷Sr hyperfine structure (*I* = 9/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
5s5p ³ P ₀	-260.0840(20)	-35.658(6)	F=7/2: +1414.12 F=9/2: +283.86 F=11/2: -1179.29
5s5p ³ P ₂	-212.7650(10)	67.215(15)	F=5/2: +2371.22 F=7/2: +1597.14 F=9/2: +618.69 F=11/2: -551.52 F=13/2: -1898.08
5s4d ³ D ₁	139.90(20)	15.0(20)	F=7/2: -762.58 F=9/2: -149.90 F=11/2: +633.30
5s4d ³ D ₃	-78.08(5)	18.0(10)	F=5/2: +867.13 F=7/2: +585.98 F=9/2: +228.99 F=11/2: -200.45 F=13/2: -698.22
5s4d ³ D ₂	-115.30(20)	51(9)	F=3/2: +1925.83 F=5/2: +1626.95 F=7/2: +1211.50 F=9/2: +682.45 F=11/2: +43.62 F=13/2: -700.30 F=15/2: -1543.80
5s4d ¹ D ₂	-5.5734(4)	55.421(6)	F=5/2: +86.71 F=7/2: +42.95 F=9/2: +0.56 F=11/2: -30.10 F=13/2: -36.30
5s5p ¹ P ₁	-3.4(4)	39(4)	F=7/2: +36.58 F=9/2: -22.60 F=11/2: -5.55
5s6s ³ S ₁	-542.00(10)	-0.1(5)	F=7/2: +2980.95 F=9/2: +542.07 F=11/2: -2439.03
5s5d ³ D ₁	227.3(7)	-	F=7/2: -1250.15 F=9/2: -227.30 F=11/2: +1022.85
5s5d ³ D ₂	-71.5(5)	-	F=5/2: +786.50 F=7/2: +536.25 F=9/2: +214.50 F=11/2: -178.75 F=13/2: -643.50
5s5d ³ D ₃	-156.9(3)	-	F=3/2: +2588.85 F=5/2: +2196.60 F=7/2: +1647.45 F=9/2: +941.40 F=11/2: +78.45 F=13/2: -941.40 F=15/2: -2118.15
5s6d ³ D ₁	239.599(11)	9.38(3)	F=7/2: -1313.49 F=9/2: -245.85 F=11/2: +1080.54
5s6d ³ D ₂	-163.2(9)	30(20)	F=5/2: +1808.95 F=7/2: +1224.62 F=9/2: +480.85 F=11/2: -416.75 F=13/2: -1461.30
5s6d ³ D ₃	-161.8(5)	20(20)	F=3/2: +2678.87 F=5/2: +2270.20 F=7/2: +1699.23 F=9/2: +967.13 F=11/2: +75.40 F=13/2: -974.13 F=15/2: -2179.30

Measured A, B constants; sources in the data file.